COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010 - Data Structures and Algorithms	**SECTION:**	CPE21S3
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HANDS-ON ACTIVITY NO. 1.1 REVIEW OF CPP PROGRAMMING

1. PROCEDURE OUTPUT

ILO A: Create Code That Follows the Basic C++ Code Structure

- Create a function that will take two numbers and display the sum.
- Create a function that will return whether variable A is greater than variable B.
- Create a function that will take two Boolean values and display the result of all logical operations then return true if it was a success.

Note:

- The driver program must call each function.
- The definitions must be after the main function.

Code + Output:

Dimaguiba_ILO_A.cpp Dimaguiba_triangleclasses.cpp

```cpp
#include <iostream>

int sumOfTwo (int fNum, int sNum);
void sumOfThree(int fNum, int sNum, int tNum);
void comparator(float aNum, float bNum);
bool logicCheck (bool a,bool b);

float aComp, bComp = 0;
bool aLogic, bLogic = 0;

int main(){


//Display the Sum part of ILO A
    std::cout<<"The sum of two numbers is:"<<sumOfTwo(6,7)<<std::endl;
    sumOfThree(6,7,8);

//Compare 2 values section of ILO A
    std::cout<<"Enter a value A (Comparison): "<<std::endl;

    while(!(std::cin>>aComp)){
        std::cout<<"Try again"<<std::endl;
        std::cin.clear();
        std::cin.ignore(10000,'\n');
    }

    std::cout<<"Enter a value B (Comparison): "<<std::endl;
    while(!(std::cin>>bComp)){
        std::cout<<"Try again"<<std::endl;
        std::cin.clear();
        std::cin.ignore(10000,'\n');
    }
    comparator(aComp,bComp);

    //Boolean
     std::cout<<"Enter a Boolean A (Comparison): "<<std::endl;

    while(!(std::cin>>aLogic)){
        std::cout<<"Try again"<<std::endl;
        std::cin.clear();
        std::cin.ignore(10000,'\n');
```

```cpp
    }

    std::cout<<"Enter a Boolean B (Comparison): "<<std::endl;
    while(!(std::cin>>bLogic)){
        std::cout<<"Try again"<<std::endl;
        std::cin.clear();
        std::cin.ignore(10000,'\n');
    }

    if (logicCheck (aLogic, bLogic)){
        std::cout<<"Success!"<<std::endl;
    }

    return 0;

}

//Definitions
int sumOfTwo (int fNum, int sNum){
    int sum = fNum + sNum;
    return sum;
}

void sumOfThree(int fNum, int sNum, int tNum){
    std::cout<<"The total sum is: "<< fNum + sNum + tNum<<std::endl;
}

void comparator(float aNum, float bNum){
    if (aNum > bNum){
        std::cout<<aNum<<" is greater than "<<bNum<<std::endl;
    }

    else if (aNum == bNum){
        std::cout<<aNum<<" is equal to "<<bNum<<std::endl;
    }

    else {
        std::cout<<bNum<<" is greater than "<<aNum<<std::endl;
    }
}
```

```cpp
bool logicCheck (bool a,bool b){
    std::cout<<"AND operation: ";
    if (a && b){
        std::cout<<"True\n";
    }

    else{
        std::cout<<"False\n";
    }

    std::cout<<"OR operation: ";
    if (a || b){
        std::cout<<"True\n";
    }

    else{
        std::cout<<"False\n";
    }

    std::cout<<"NOT operation:\n";
    std::cout<<"NOT A = "<<!a<<std::endl;
    std::cout<<"NOT B = "<<!b<<std::endl;

    return true;


}
```

```
The sum of two numbers is:13
The total sum is: 21
Enter a value A (Comparison):
51
Enter a value B (Comparison):
23
51 is greater than 23
Enter a Boolean A (Comparison):
1
Enter a Boolean B (Comparison):
0
AND operation: False
OR operation: True
NOT operation:
NOT A = 0
NOT B = 1
Success!

--------------------------------
Process exited after 13.16 seconds with return value 0
```

ILO B: Implement Appropriate Class Definition and Instances Based on Given Requirements

Code + Output:

```cpp
#include <iostream>

//PascalCase
class Triangle{
    private:
        double totalAngle, angleA, angleB, angleC;
    public:
        //constructor
        Triangle(double A, double B, double C);
        void setAngles(double A, double B, double C);
        const bool validateTriangle();
};

int main(){

    Triangle set1(40,30,110);

    if(set1.validateTriangle()){
        std::cout<<"The shape is a valid triangle\n";
    }
    else{
        std::cout<<"The shape is NOT a valid triangle\n";
    }
}

Triangle::Triangle(double A, double B, double C){
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A + B + C;
}

void Triangle::setAngles(double A, double B, double C){
    angleA = A;
    angleB = B;
    angleC = C;
}
```

```cpp

const bool Triangle::validateTriangle(){
    return (totalAngle <= 180);
}
```

```
The shape is a valid triangle

--------------------------------
Process exited after 1.006 seconds with return value 0
Press any key to continue . . .
```

Output Section:

Sections	Answer
Header File Declaration Section	`#include <iostream>`

Global Declaration Section	`float aComp, bComp = 0;`
	`bool aLogic, bLogic = 0;`
	(taken from ILO-A)

Class Declaration and Method Definition Section	`class Triangle{`
	`  private:`
	`  double totalAngle, angleA, angleB, angleC;`
	`  public:`
	`  //constructor`
	`  Triangle(double A, double B, double C);`
	`  void setAngles(double A, double B, double C);`
	`  const bool validateTriangle();`
	`};`

Main Function	`int main(){`
	`  Triangle set1(40,30,110);`
	`  if(set1.validateTriangle()){`
	`  std::cout<<"The shape is a valid triangle\n";`
	`  }`
	`  else{`
	`  std::cout<<"The shape is NOT a valid triangle\n";`
	`  }`
	`}`

Method Definition	`Triangle::Triangle(double A, double B, double C){`
	`  angleA = A;`
	`  angleB = B;`
	`  angleC = C;`
	`  totalAngle = A + B + C;`
	`}`
	`void Triangle::setAngles(double A, double B, double C){`
	`  angleA = A;`
	`  angleB = B;`
	`  angleC = C;`
	`}`
	`const bool Triangle::validateTriangle(){`
	`  return (totalAngle <= 180);`
	`}`

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

1. Create a C++ program to swap the two numbers in different variables.

Code + Output:

```cpp
#include <iostream>

void swapper(int aNum, int bNum);

int main(){
    int funnyNumber1, funnyNumber2, temp = 0;

    std::cout<<"Number Swapper:\n\n";
    std::cout<<"Enter the FIRST number, ANY NUMBER YOU PLEASE: "; //Flavor text, pls dont sue

    while(!(std::cin>>funnyNumber1)){
        std::cout<<"I SAID NUMBER. TRY. AGAIN.\n";
        std::cin.clear();
        std::cin.ignore(10000,'\n');
    }

    std::cout<<"Enter the SECOND number, ANY NUMBER YOU PLEASE: ";

    while(!(std::cin>>funnyNumber2)){
        std::cout<<"I SAID NUMBER. TRY. AGAIN.\n";
        std::cin.clear();
        std::cin.ignore(10000,'\n');
    }

    swapper(funnyNumber1, funnyNumber2);

    return 0;
}

void swapper(int aNum, int bNum){

    std::cout<<"Your current number order: "<<aNum<<" "<<bNum<<std::endl;
    int temp = aNum;
    aNum = bNum;
    bNum = temp;
    std::cout<<"SWAPPED: "<<aNum<<" "<<bNum<<std::endl;

}
```

```
Number Swapper:

Enter the FIRST number, ANY NUMBER YOU PLEASE: f
I SAID NUMBER. TRY. AGAIN.
7
Enter the SECOND number, ANY NUMBER YOU PLEASE: 6
Your current number order: 7 6
SWAPPED: 6 7

--------------------------------
Process exited after 8.086 seconds with return value 0
Press any key to continue . . .
```

Analysis:

The code makes use of a swap function. In this swap function, an extra variable is utilized. A temporary variable or "temp" in the code, holds on to the value of one variable and gives it to the second variable after the first copies it.

2. Create a C++ program that has a function to convert temperature in Kelvin to Fahrenheit.

Code + Output:

```cpp
#include <iostream>
#include <stdlib.h> //to clear screen

float kelToFa(float kel);
float faToKel(float fah);

int main(){
    int selection = 0;
    float temperature= 0;
    int exitVal = 0;

    while(exitVal != 1){

    std::cout<<"Menu:\n1. Kelvin to Fahrenheit\n2. Fahrenheit to Kelvin\n3.Quit\n";
    std::cout<<"Select: ";

    while(!(std::cin>>selection)){
        std::cout<<"Try again\n "<<std::endl;
        std::cin.clear();
        std::cin.ignore(10000,'\n');
    }

    switch(selection){


        case 1:
            std::cout<<"Enter temperature(Kelvin): ";
            while(!(std::cin>>temperature)){
                std::cout<<"Try again\n "<<std::endl;
                std::cin.clear();
                std::cin.ignore(10000,'\n');
            }
            std::cout<<kelToFa(temperature)<<" degrees Fahrenheit\n\n"<<std::endl;
            std::cout<<"Press Enter to Continue\n";
            std::cin.get();
            std::cin.ignore(10000,'\n');
            system("cls");
            break;

        case 2:
            std::cout<<"Enter temperature(Fahrenheit): ";
            while(!(std::cin>>temperature)){
                std::cout<<"Try again\n "<<std::endl;
                std::cin.clear();
                std::cin.ignore(10000,'\n');
            }
            std::cout<<faToKel(temperature)<<" degrees Kelvin\n\n"<<std::endl;
            std::cout<<"Press Enter to Continue\n";
            std::cin.get();
            std::cin.ignore(10000,'\n');
            system("cls");
            break;

        case 3:
            exitVal = 1;
            std::cout<<"Quitting"<<std::endl;
            break;

        default:
            std::cout<<"Invalid Choice!\n";
            break;
        }

}
    return 0;
}

float kelToFa(float kel){
    float solution = (kel * 1.8) - 459.67;
    return solution;
}

float faToKel(float fah){
    float solution = (fah + 459.67) * (5.0/9.0);
    return solution;
}
```

```
Menu:
1. Kelvin to Fahrenheit
2. Fahrenheit to Kelvin
3.Quit
Select: 1
Enter temperature(Kelvin): 21
-421.87 degrees Fahrenheit


Press Enter to Continue
```

```
Menu:
1. Kelvin to Fahrenheit
2. Fahrenheit to Kelvin
3.Quit
Select: 2
Enter temperature(Fahrenheit): 500
533.15 degrees Kelvin


Press Enter to Continue
```

```
Menu:
1. Kelvin to Fahrenheit
2. Fahrenheit to Kelvin
3.Quit
Select: 3
Quitting

--------------------------------
Process exited after 69.97 seconds with return value 0
Press any key to continue . . .
```

Analysis:

The code uses 2 functions. One from Kelvin to Fahrenheit and vice-versa. A switch-case loop is utilized to return to the main menu after an operation. The screen clears once the user presses ENTER and quits when they select 3.

3. Create a C++ program that has a function that will calculate the distance between two points.

Code + Output:

```cpp
#include <iostream>
#include <cmath> // for squareroot

float distanceCalculator(float x1, float x2, float y1, float y2);

int main(){
    float x1,x2,y1,y2 = 0;

    std::cout<<"Enter first point (X): "<<std::endl;

    while(!(std::cin>>x1)){
        std::cout<<"Try again"<<std::endl;
        std::cin.clear();
        std::cin.ignore(10000,'\n');
    }

    std::cout<<"Enter second point (X): "<<std::endl;

    while(!(std::cin>>x2)){
        std::cout<<"Try again"<<std::endl;
        std::cin.clear();
        std::cin.ignore(10000,'\n');
    }

    std::cout<<"Enter first point (Y): "<<std::endl;

    while(!(std::cin>>y1)){
        std::cout<<"Try again"<<std::endl;
        std::cin.clear();
        std::cin.ignore(10000,'\n');
    }

    std::cout<<"Enter second point (Y): "<<std::endl;

    while(!(std::cin>>y2)){
        std::cout<<"Try again"<<std::endl;
        std::cin.clear();
        std::cin.ignore(10000,'\n');
    }

    std::cout<<distanceCalculator(x1,x2,y1,y2)<<" units";

    return 0;
}

float distanceCalculator(float x1, float x2, float y1, float y2){
    float solution = std::sqrt( pow((x2-x1),2) + pow((y2-y1),2) );
    return solution;
}
```

```
Enter first point (X):
1
Enter second point (X):
4
Enter first point (Y):
2
Enter second point (Y):
6
5 units
--------------------------------
Process exited after 57.73 seconds with return value 0
Press any key to continue . . .
```

Analysis:

Using cmath to allow the program to use mathematical operations such as the square root and the power function. Following the formula and inputting the 2 points into the function, the final answer is found.

4. Modify the code given in ILO B and add the following functions:

a. A function to compute for the area of a triangle

b. A function to compute for the perimeter of a triangle

c. A function that determines whether the triangle is acute-angled, obtuse-angled or 'others.'

Modifications + Output:

```cpp
void setArea(double bs, double ht);
double displayArea();
double perimeter(double side1, double side2, double side3);
void triType();
```

```cpp
Triangle::Triangle(double A, double B, double C, double bs, double ht){
    angleA = A;
    angleB = B;
    angleC = C;
    base = bs;
    height = ht;
    totalAngle = A + B + C;
    area = (0.5 * (base * height));
}
```

```cpp
void Triangle::setArea(double bs, double ht){
    base = bs;
    height = ht;
}
double Triangle::displayArea(){
    return area;
}
double Triangle::perimeter(double side1, double side2, double side3){
    return side1 + side2 + side3;
}

void Triangle::triType(){
    if (angleA == 90 || angleB == 90 || angleC == 90 ){ //if either has 90 degrees
        std::cout<<"A Right Triangle"<<std::endl;
    }
    else if (angleA > 90 || angleB > 90 || angleC > 90){
        std::cout<<"An Obtuse Triangle"<<std::endl; //when one is greater than 90
    }
    else{
        std::cout<<"An Acute Triangle"<<std::endl; // all are less than 90
    }
}
```

```
Area: 6 units

Perimeter: 12 units

The shape is a valid triangle
An Obtuse Triangle

--------------------------------
Process exited after 0.8671 seconds with return value 0
Press any key to continue . . .
```

Analysis:
In addition to the code in the first place, a few changes were added on the private section of the class, in that the base and the height was added. In the public section, a function relating to the perimeter was also added. The type of triangle was also added through the same means. Since they are all in one class, they may share variables and values within the class. Also, although optional, additional functions including setting new values for the area are also added. All of which are then called upon under the main line of code leaving us with an output.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: *ChatGPT*

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: Explanation of concepts

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: Mainly used in figuring out how classes interact with themselves, once figured out, additional features and functions added were purely by human choice.

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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